

IoMT: Internet of Medical Things

An Autonomous Edge-AI Patient Monitoring System

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1 Abstract

In modern healthcare facilities, the ratio of medical staff to patients often prevents continuous, 24x7 direct observation. This gap in monitoring can lead to delayed detection of critical physiological changes, resulting in preventable complications or fatalities. This project introduces an Internet of Medical Things (IoMT) solution: an intelligent edge device equipped with multimodal sensors (ECG, Accelerometer, Visual) and a centralized AI dashboard. By leveraging a 3D-CNN for arrhythmia detection, a Multi-Task Neural Network for EHR risk scoring, and a visual gesture recognition agent, the system provides real-time abnormality detection and instant alerting, aiming to reduce mortality rates in post-surgery and intensive care units.

2 Introduction

2.1 Problem Statement

Nurses and doctors face significant challenges in observing every patient continuously. A yearly review at AIG Hospital highlighted the severity of this issue, revealing that 72 post-surgery patients deteriorated due to insufficient monitoring, with 6 cases resulted in fatalities. Manual monitoring is prone to human error and fatigue, and current automated systems often suffer from high rates of false positives (e.g., bed alarms triggered by shifting weight).

2.2 Objective

The primary objective of this project is to create an autonomous "Edge Triage Agent" that:

1. Monitors patient vitals and physical state in real-time.
2. Uses multimodal AI to validate anomalies before alerting staff, thereby reducing alarm fatigue.
3. Predicts clinical risks using historical Electronic Health Records (EHR).

3 System Architecture and Data Flow

The system relies on a robust IoT pipeline designed for speed and reliability.

3.1 Hardware Layer

The edge node consists of:

- **Sensors:** Electrocardiogram (ECG) leads for heart monitoring and accelerometers for motion tracking. The choice of sensors allows for the capture of both vital signs and physical activity, which are critical inputs for the multimodal AI.

- **Visual Input:** ESP32-CAM module for computer vision tasks, providing contextual validation for physical events like falls or distress signals.
- **Processing Unit:** An edge computing unit running Arduino-based firmware to collect raw sensor data. Processing at the edge ensures low latency for time-critical monitoring tasks.

3.2 Communication & Database Layer

To handle the high-velocity data stream characteristic of real-time monitoring, we implemented a Publish-Subscribe architecture:

1. **MQTT Server:** Acts as the central broker. The sensors publish readings to specific topics in JSON format. This lightweight protocol ensures minimal overhead and low latency, crucial for IoMT applications.
2. **Telegraf (Subscriber Node):** A server-side agent that subscribes to the MQTT topics. It handles data buffering, formatting, and reliable ingestion into the database.
3. **InfluxDB:** A time-series database chosen for its high write throughput and query speed. It efficiently stores sensor data, optimizing it for time-based analysis, model inference, and real-time visualization on the AI dashboard.

4 Methodology and AI Modalities

The intelligence of the system is divided into three distinct modalities, each targeting a specific aspect of patient health and risk.

4.1 Modality 1: Arrhythmia Detection

Architecture: We implemented a 3D-CNN architecture utilizing 1D Convolutional layers. The network progresses through filters of size $32 \rightarrow 64 \rightarrow 128$, incorporating Batch Normalization and Dropout layers to prevent overfitting. The model contains over 140,000 trainable parameters, allowing it to capture subtle, high-frequency anomalies in the ECG signal.

Training Strategy:

- **Data Augmentation:** The model was trained on a combination of public online datasets (e.g., MIT-BIH) and custom-collected samples. To simulate arrhythmia conditions safely, team members performed strenuous exercises (jumping jacks, high knees) while wearing sensors to generate varied, non-standard heart rhythms.
- **Optimization:** A class-balanced weighting scheme was used to handle data imbalance (since normal beats heavily outnumber arrhythmic events). Early stopping was implemented with a patience of 3 epochs on Validation AUC to maximize generalization.

- **Performance:** The model achieved a Validation AUC of $\sim 90\%$ and a Testing AUC of **93.36%**, indicating high reliability in distinguishing normal and abnormal heart activity.

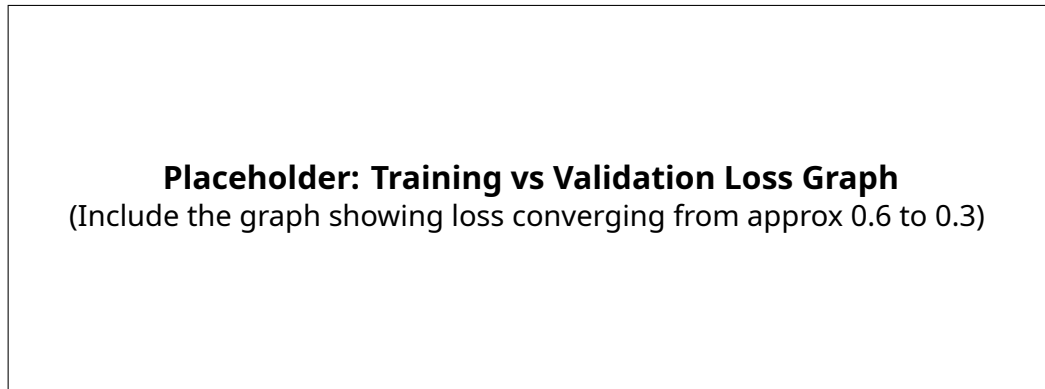


Figure 1: Training vs Validation Loss for Arrhythmia Model

4.2 Modality 2: EHR Risk Prediction

Data Scale: The model was trained on over 10 million individual records derived from FHIR JSON files, covering patient demographics, lab results, medications, and diagnoses. This large scale ensures the model captures a broad spectrum of clinical patterns.

Feature Engineering: We utilized parallel processing to clean raw data and extract 130+ key features. Advanced engineering techniques included:

- 3-hour rolling averages to detect physiological trends, providing more context than single point-in-time measurements.
- Delta values to measure rapid changes in vitals, which often signify acute deterioration.

Model Architecture (Multi-Task Network):

- **Shared Body:** Deep Dense layers ($64 \rightarrow 32$) that learn general, underlying representations of patient health ("sickness patterns") applicable across various complications.
- **Output Heads:** 153 separate output layers. Each head predicts a specific clinical risk (e.g., Sepsis in 6h, Cardiac Arrest in 24h) based on the features learned by the shared body, allowing for granular and proactive intervention planning.

4.3 Modality 3: Gesture Recognition & Autonomous Triage

The Problem: False positives in hospitals are rampant. Bed alarms trigger when patients shift weight, and fall detectors trigger when items drop. This leads to severe alarm fatigue among nursing staff.

The Solution: An autonomous agent that validates motion visually before escalating an alert. The system integrates the three modalities into a single metric:

$$\text{Final Score} = (\text{Medical Risk}) + (\text{Physical Event}) + (\text{Visual Confirmation}) \quad (1)$$

The system only alerts the nurse if the Weighted Fusion Score exceeds a threshold, confirming that a physiological or physical distress signal is genuine and requires immediate attention.

5 Challenges and Solutions

Overfitting: Initial models memorized training data instead of generalizing, resulting in poor performance on unseen data. *Solution:* Implemented class weight balancing for imbalanced datasets and switched metric monitoring from pure accuracy to Area Under the Curve (AUC), a more robust measure of model performance on skewed classes.

Sensor Noise: Low-cost sensors provided inconsistent readings compared to medical-grade equipment, introducing noise into the data stream. *Solution:* Rigorous calibration protocols based on manufacturer documentation were established, coupled with digital signal processing (DSP) filters implemented in the Arduino firmware.

Big Data Constraints: The 10M+ row EHR dataset exceeded conventional server RAM capacity, hindering rapid prototyping. *Solution:* Implemented a streaming data generator using `tf.data`. This allows batch loading (e.g., Batch Size = 8) directly from disk storage during training, efficiently utilizing memory without sacrificing dataset size.

6 Future Scope: Hospital Integration

To transition this prototype into a deployment-ready, regulatory-compliant hospital system, the following critical developments are proposed, focusing on data security, scalability, and workflow integration.

6.1 Integration with Hospital Information Systems (HIS)

The current AI dashboard must be fully integrated into the hospital's central clinical workflow.

- **Protocol Compliance:** Developing connectors that adhere to standard interoperability protocols like **HL7 (Health Level Seven)** and **FHIR (Fast Healthcare Interoperability Resources)**.
- **Workflow Seamlessness:** Alerts generated by our Edge/AI system could automatically populate in the central Nursing Station (CNS) software or dispatch systems, ensuring staff receive critical notifications through existing, familiar channels, thereby achieving seamless workflow integration.

6.2 Federated Learning for Privacy

Patient data privacy (HIPAA/GDPR) is paramount in healthcare environments.

- **Decentralized Training:** Future iterations will implement **Federated Learning**. Models are trained locally on edge devices within the hospital network.
- **Anonymity Preservation:** Only model weight updates (not raw patient data) are aggregated by the central server. This preserves patient anonymity while leveraging the collective data to continuously improve the model's accuracy across different demographics and hospital units.

6.3 Predictive Maintenance & Scalability

Scaling the deployment across an entire hospital requires a robust and maintainable network architecture.

- **Mesh Networking:** Implementing a low-power, wide-area mesh network topology (e.g., Zigbee or LoRaWAN) would allow hundreds of these edge devices to communicate efficiently without clogging the hospital's primary high-bandwidth Wi-Fi network.
- **Self-Diagnosis:** Integrating self-diagnostic routines into the edge devices is crucial. This feature would allow the device to alert technicians if a sensor (like an ECG lead) degrades, disconnects, or is providing statistically anomalous noise, enabling proactive maintenance.

6.4 Post-Discharge Home Monitoring

The device's compact, portable design makes it suitable for extending clinical care beyond the hospital walls.

- **Hospital-at-Home:** The system can be deployed in a "Hospital-at-Home" setting. Patients discharged after critical procedures or complex surgeries can wear the device at home.
- **Telemedicine Alerting:** The device would continue to monitor for risks such as arrhythmia recurrence or falls, automatically alerting the hospital's telemedicine wing or remote monitoring center if a defined threshold of clinical deterioration is detected, facilitating rapid remote intervention.

7 Conclusion

The IoMT project by Team AJAS demonstrates a viable proof-of-concept for autonomous patient monitoring. By fusing physiological data (ECG) with environmental context (Computer Vision) and historical risk (EHR), the system achieves a high degree of reliability (93% AUC). This solution addresses the critical need for continuous observation, offering a scalable, reliable path to reducing preventable hospital fatalities and enhancing the efficiency of clinical staff.